Data Management Mini Toolkit (DMMT)

School of Engineering & Technology (SOET)

Program: B. Tech CSE (AI & ML), 2nd SEM

Course Title: Data Structure

Course Code: ETCCDS202

Unit Title: Non-Linear & Advanced Data Structures

Submitted by: Samarth Sharma

Roll no: 2501730279

Section: A

Submission Date: April 2026

INTRODUCTION

In this assignment, I have implemented three important data structures in Python:

- Binary Search Tree (BST)
- Graph (using BFS and DFS)
- Hash Table (with collision handling)

These data structures are widely used in real-world applications like databases, searching systems, and networks.

Binary Search Tree (BST)

What I did?

I created a BST where I can:

- Insert elements
- Search elements
- Delete elements
- Print inorder traversal

Why inorder gives sorted output?

In BST, left side always has smaller values and right side has larger values.
So when we follow inorder (Left ---> Root ---> Right), the output automatically comes in sorted order.

Time Complexity:

- Average case: O(log n)
- Worst case: O(n) (when tree becomes like a linked list)

Graph (BFS and DFS)

What I did?

I created a directed and weighted graph using adjacency list.
Then I implemented:

- BFS (Breadth First Search)
- DFS (Depth First Search)

Difference between BFS and DFS:

- BFS goes level by level
- DFS goes deep into one branch first

Real-life use:

- BFS: used for finding shortest path
- DFS: used in checking cycles or connectivity

Hash Table

What I did?

I implemented a hash table using separate chaining.
Operations performed:

- Insert
- Get
- Delete

What is collision?

Collision happens when two keys get same index in hash table.

Why separate chaining is used?

To handle collision, multiple elements are stored in same bucket using a list.

Time Complexity:

- Average case: O(1)
- Worst case: O(n)

Observations

- BST helps in storing data in sorted form
- Graph helps in understanding connections between nodes
- Hash table is very fast for searching

Conclusion

From this assignment, I understood how different data structures work and why they are important. It also helped me understand real-world applications of these concepts.

References

- GeeksforGeeks – Data Structures Concepts https://www.geeksforgeeks.org
- Python Official Documentation https://docs.python.org/3/